

Quiz 5

● Graded

Student

Samyak Sanghvi

Total Points

1 / 4 pts

Question 1

Q1

1 / 2 pts

✓ + 0.5 pts Mean: no

✓ + 0.5 pts Mean justification: don't know the original intensities of the original image

+ 0.5 pts Variance: yes

+ 0.5 pts Variance justification: take variance of intensities of pixels in constant regions

+ 0 pts No/Wrong answer

Question 2

Q2

0 / 2 pts

+ 1 pt Wiener filter formula: $W = H^* / (|H|^2 + S_n/S_f)$

+ 2 pts $W = 0$ when $H = 0$

+ 0.25 pts (Optional) noise is also zeroed out due to weiner

✓ + 0 pts No/Wrong answer

Name: Gamyak Sanghvi

COL7(6)83 Quiz 5
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1. You are given an image g degraded by both additive noise and blur. The only additional knowledge you have is that you can identify which regions should have constant intensity and which should contain texture or other intensity variations. Can you determine (i) the mean, and (ii) the variance of the additive noise? If so, how; if not, why not?

(i)

$$g = (f + n)^* w$$

(i) no since I can't figure out what avg f value in regions of const. intensity will be, we can get

$$\bar{g} = (\bar{f} + \bar{n})^* w$$

we can't get $\bar{n}$

(ii) since I don't have the mean, I can't find the variance

2. A fundamental barrier in deconvolution is that at frequencies (u, v) where $H(u, v) = 0$, information about the original image is completely lost. What is the behaviour of Wiener filtering at such frequencies? Assume both the image and the noise have nonzero spectra.